EXECUTIVE SUMMARY

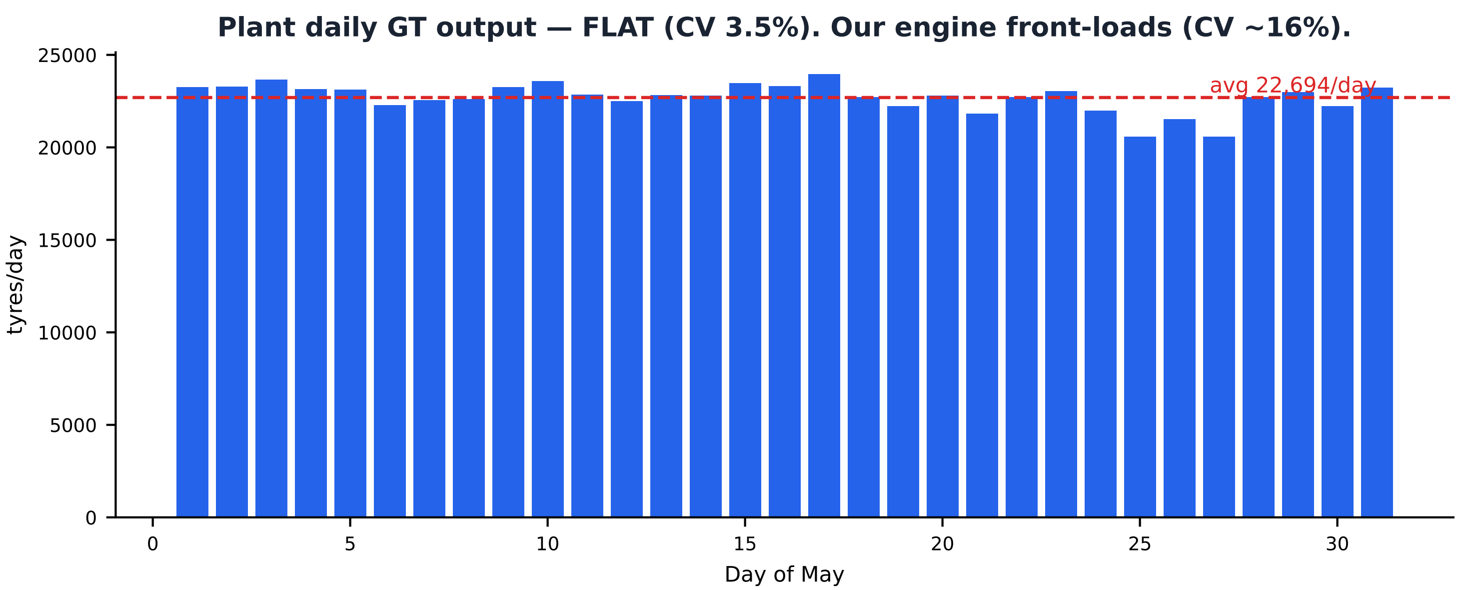
How the plant makes 703k in May

703,509	671,138	~32,000
Plant output (101% of demand)	Our engine (96.7%)	Gap to close

THE ONE FINDING

The real plant does NOT follow a 5-day size-change rule.

It switches a machine's size whenever a press needs it — median gap only ~7 hours.



What this means

- The plant nearly meets demand (703k of 694k) by keeping every press fed with the right size at the right moment.
- It does this with many short, responsive campaigns and a flat daily output — not by dwelling on a size.
- Our engine's 5-day dwell/cooldown rule is the single biggest thing holding output below the plant.

The plant changes size in hours, not days

How many different-size changeovers per month

How we count a changeover:

the machine ran a REAL batch (≥40 tyres) of a new size. Stray 1–2 tyre blips are ignored as sensor noise.

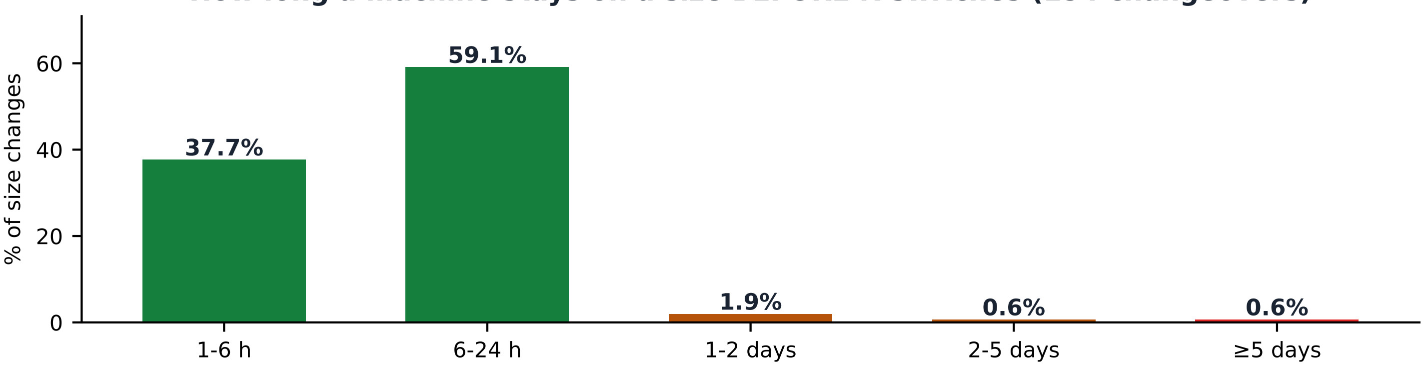
Configuration	COs/mo	/machine	
Real plant (May)	**154**	4.4	the clean, real count
Our engine — adopted 5-day rule	**75**	2.1	half the plant → we switch too little
Our engine — free-bounce (no rule)	**293**	8.4	more than plant → uncontrolled

Those 154 size changes, split by TYPE

Demand-complete (green): 7 — machine finished that size, never returned. The ONLY type our rule allows anytime.

JIT excursions (red): 147 — machine left a size and RETURNED later (not complete). Exactly what our 5-day rule blocks.

How long a machine stays on a size BEFORE it switches (154 changeovers)



5-DAY-RULE VERDICT: NO — the plant does not dwell.

Median gap ~7 hours • ~97% of switches within 1 day • 95% are JIT excursions our rule blocks

Most churn is CHEAP same-size switches, not size changes

- Same-size SKU changeovers: 1298 vs diff-size changeovers: 154 — about a 8:1 ratio.
- Same-size COs are cheap (20-60 min on VMI/BJ); diff-size COs are the costly ones (88-180 min) — the plant keeps those moderate.

CONCLUSIONS

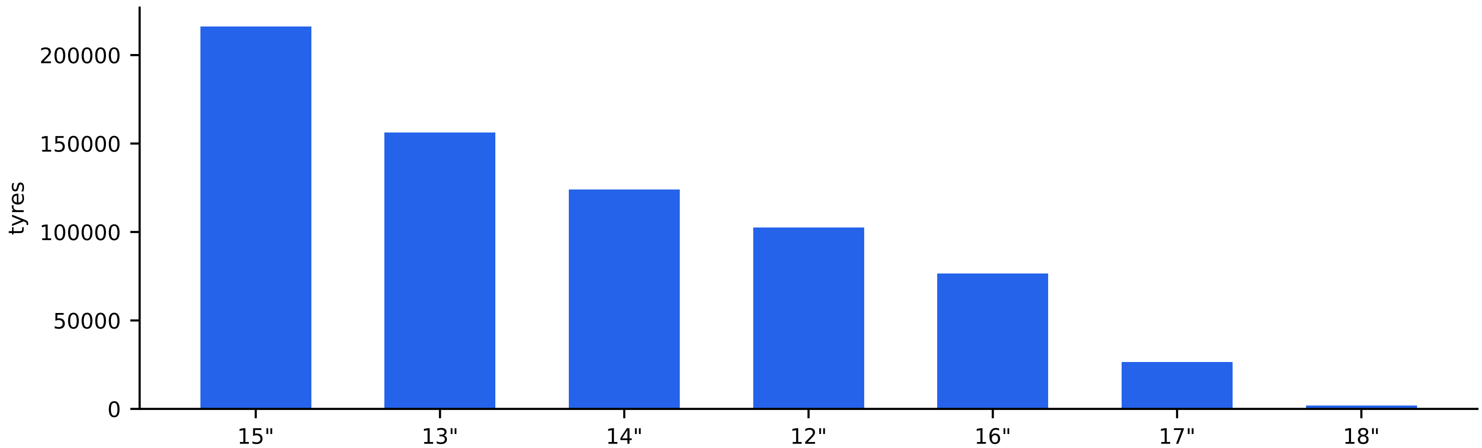
What to change to reach plant-like output

KEEP ✓

Stage-1 single-inch rule.
Plant keeps it too (3 of 22 S-1 machines change size all month).

DROP ✗

The 5-day dwell / cooldown rule.
Plant data contradicts it — it switches size in hours.



Root causes of our ~32k gap

- TIMING, not count. Plant does 154 diff-COs (we do 75) but switches JIT (median 7 h). Our 5-day dwell blocks the JIT switch a press needs.
- FLAT vs front-loaded. Plant CV 3.5%; we run ~16% — we build too much early, starving presses late in the month.
- CONCURRENCY. Plant builds 6-7 sizes at once every day; our band + dwell rules narrow how many sizes are live together.
- Curing scarcity on 13"/15" is the residual floor — real capacity limit, not scheduling (unchanged by the above).

Recommendations (in priority order)

1. Replace the 5-day rule with a JIT size-switch trigger bounded by a per-machine per-day CO budget. Our own free test already shows +17,205 cured on May.
2. Add a level-loading / pacing lever so building is flat across the month (match the plant's flat curve), not front-loaded.
3. Widen VMI / BJ inch flexibility so more sizes can be live at once.
4. Keep Stage-1 single-inch (validated) and accept the trade-off: many more COs, but mostly cheap same-size ones.

What size each machine runs — REAL PLANT

Most machines run 1-2 sizes; 8 flex across 2-3 (amber). Sizes listed most-produced first.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	**13, 15"**
6803	bj3stage1	15"
6909	nrm9stage1	12"
6911	nrm11stage1	**15, 13"**
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	12"
7802	midland2stage1	12"
7803	midland3stage1	15"
7804	midland4stage1	**15, 14"**
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	15"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	**12, 15"**
8302	gtic2	**12, 13"**
8501	vmi1	**15, 13"**
8502	vmi2	**15, 14"**

Unistage

code	plant name	sizes
6001	VMIExxium01	14"
6002	VMIExxium02	15"
6003	VMIExxium03	17"
6004	VMIExxium04	16"
7001	vmi1Maxx	**16, 15, 18"**
7002	vmi2Maxx	14"
7003	vmi3Maxx	15"
7004	vmi4Maxx	14"
7101	bj4	15"
7102	bj5	15"
7103	bj6	13"
7104	bj7	15"
7105	bj9	13"
7106	bj10	13"
7201	bj8	16"
7501	us1	12"
7502	us2	13"
7503	us3	13"

KEY READ

The plant runs machines on 1-2 sizes each — FOCUSED — but switches among them FAST (154 changeovers, ~7 h dwell). Its edge is timing, not a wide size range per machine.

690,835 cured · 88.6% coverage · 127 diff-size COs

What size each machine runs under the adopted July config (15 machines multi-inch, amber). JIT + hybrid re-anchoring.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	15"
6803	bj3stage1	15"
6909	nrm9stage1	12"
6911	nrm11stage1	15"
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	14"
7802	midland2stage1	13"
7803	midland3stage1	12"
7804	midland4stage1	15"
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	15"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	12"
8302	gtic2	**12, 15, 13"**
8501	vmi1	**15, 13"**
8502	vmi2	**15, 16, 14"**

Unistage

code	plant name	sizes
6001	VMIExxium01	14"
6002	VMIExxium02	**14, 15"**
6003	VMIExxium03	**16, 17, 15"**
6004	VMIExxium04	**16, 15, 17, 14"**
7001	vmi1Maxx	**15, 16, 14"**
7002	vmi2Maxx	**18, 14, 16, 17"**
7003	vmi3Maxx	**17, 15, 16, 14"**
7004	vmi4Maxx	**14, 15, 16"**
7101	bj4	15"
7102	bj5	15"
7103	bj6	**13, 15"**
7104	bj7	15"
7105	bj9	**13, 15"**
7106	bj10	**13, 15"**
7201	bj8	**16, 13"**
7501	us1	**13, 12"**
7502	us2	13"
7503	us3	13"

KEY READ

Adopted July config (CO limit 10). 690,835 cured / 127 diff-size COs — the balanced knee: near-peak KPI with plant-like churn. Machines concentrate on a dominant inch; only the amber ones flex across 2-3.

APPENDIX A3 — OUR JULY PLAN (margin=200, budget=2)

683,536 cured · 87.7% coverage · 83 diff-size COs

Leaner-churn alternative (14 machines multi-inch, amber). Higher urgency margin = fewer size switches.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	15"
6803	bj3stage1	15"
6909	nrm9stage1	12"
6911	nrm11stage1	15"
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	(idle)"
7802	midland2stage1	13"
7803	midland3stage1	12"
7804	midland4stage1	15"
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	15"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	12"
8302	gtic2	**12, 13, 15"**
8501	vmi1	**15, 13"**
8502	vmi2	**15, 16"**

Unistage

code	plant name	sizes
6001	VMIExxium01	**14, 15"**
6002	VMIExxium02	**16, 14, 15, 17"**
6003	VMIExxium03	17"
6004	VMIExxium04	**16, 14"**
7001	vmi1Maxx	**14, 16"**
7002	vmi2Maxx	**18, 16, 14, 17"**
7003	vmi3Maxx	15"
7004	vmi4Maxx	**15, 14, 16"**
7101	bj4	15"
7102	bj5	15"
7103	bj6	**13, 15"**
7104	bj7	15"
7105	bj9	**13, 15"**
7106	bj10	**13, 15"**
7201	bj8	**16, 13"**
7501	us1	**13, 12"**
7502	us2	13"
7503	us3	13"

KEY READ

Lower-churn July alternative: only 83 diff-size COs (vs 127) but ~7.3k less cured. Raising the urgency margin 100→200 trims switching but costs coverage — use only if minimising changeovers matters more than KPI.

The adopted approach (validated on May/June/July)

1. Replaced the 5-day rule with JIT switching + a daily CO budget

- **Trigger (JIT)**
  A machine changes size the moment a curing press needs a size it has no GT for — no dwell, no cooldown. This unlocks the switches the plant makes (median ~7 h).
- **Control (no bounce to 293)**
  An urgency margin (only switch if the target inch is meaningfully more starving) + a per-machine per-day diff-CO budget. Holds churn to ~100-150 diff-COs while keeping the KPI.
- **Kept**
  Demand-complete immediate switch, and Stage-1 single-inch (validated — the plant keeps it too).

Result vs the 5-day dwell rule (each month, own demand + moulds + building data):

May 671,138→685,002 · June 610,159→644,803 · July 640,578→690,835 — all mould-feasible.

2. Seeded each month from its real building-running data + demand re-anchoring

- **Hybrid initial allocation**
  Each machine keeps its real running size when that size has demand this month; otherwise it is re-anchored to the neediest inch. Fully demand-dynamic (5-6 machines re-anchored per month).
- **Month-specific inputs**
  Demand file, running-moulds table, and building-running snapshot per month; curing CO limit 12 for May/June, 10 for July (from the CO-limit sweep).

BOTTOM LINE

Dropping the dwell and switching size JIT — controlled by urgency margin + daily budget — reaches plant-like output on every month, with churn tunable via the margin (higher = fewer diff-size COs, small KPI cost).

679,962 cured · 98.0% coverage · 62 diff-size COs

What size each machine runs under the adopted May config (13 machines multi-inch, amber). JIT + hybrid re-anchoring.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	15"
6803	bj3stage1	(idle)"
6909	nrm9stage1	12"
6911	nrm11stage1	15"
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	12"
7802	midland2stage1	16"
7803	midland3stage1	(idle)"
7804	midland4stage1	(idle)"
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	15"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	12"
8302	gtic2	**12, 15"**
8501	vmi1	**13, 15"**
8502	vmi2	16"

Unistage

code	plant name	sizes
6001	VMIExxium01	14"
6002	VMIExxium02	**15, 14"**
6003	VMIExxium03	**17, 16, 15"**
6004	VMIExxium04	**16, 14"**
7001	vmi1Maxx	**15, 16, 14"**
7002	vmi2Maxx	**14, 18"**
7003	vmi3Maxx	**16, 15, 17"**
7004	vmi4Maxx	**14, 17, 15"**
7101	bj4	15"
7102	bj5	15"
7103	bj6	**13, 15, 16"**
7104	bj7	**15, 16"**
7105	bj9	13"
7106	bj10	13"
7201	bj8	**13, 16"**
7501	us1	**13, 12"**
7502	us2	13"
7503	us3	13"

KEY READ

Adopted May config (CO limit 12, demand_may 693,748, Daily_Running_Moulds). 679,962 cured / 62 diff-size COs — near-full coverage with plant-like churn. Machines concentrate on a dominant inch; only the amber ones flex across 2-3.

APPENDIX A5 — OUR MAY PLAN (margin=200, budget=2)

667,307 cured · 96.2% coverage · 41 diff-size COs

Leaner-churn May alternative (10 machines multi-inch, amber). Higher urgency margin = fewer size switches.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	(idle)"
6803	bj3stage1	(idle)"
6909	nrm9stage1	12"
6911	nrm11stage1	15"
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	12"
7802	midland2stage1	16"
7803	midland3stage1	(idle)"
7804	midland4stage1	(idle)"
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	15"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	12"
8302	gtic2	**12, 15, 13"**
8501	vmi1	13"
8502	vmi2	16"

Unistage

code	plant name	sizes
6001	VMIExxium01	14"
6002	VMIExxium02	**15, 17"**
6003	VMIExxium03	**17, 15"**
6004	VMIExxium04	**14, 16"**
7001	vmi1Maxx	**15, 16, 14"**
7002	vmi2Maxx	14"
7003	vmi3Maxx	**16, 15, 17"**
7004	vmi4Maxx	**16, 14, 17"**
7101	bj4	15"
7102	bj5	15"
7103	bj6	13"
7104	bj7	**15, 16"**
7105	bj9	13"
7106	bj10	**15, 13"**
7201	bj8	**13, 14"**
7501	us1	12"
7502	us2	13"
7503	us3	13"

KEY READ

Lower-churn May alternative: only 41 diff-size COs (vs 62) but ~12.7k less cured. Raising the urgency margin 150->200 trims switching but costs coverage — use only if minimising changeovers matters more than KPI.

643,928 cured · 98.1% coverage · 103 diff-size COs

What size each machine runs under the adopted June config (17 machines multi-inch, amber). JIT + hybrid re-anchoring.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	(idle)"
6803	bj3stage1	16"
6909	nrm9stage1	12"
6911	nrm11stage1	15"
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	12"
7802	midland2stage1	13"
7803	midland3stage1	15"
7804	midland4stage1	15"
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	(idle)"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	**12, 13"**
8302	gtic2	**12, 15, 13"**
8501	vmi1	**15, 12, 13"**
8502	vmi2	**16, 15"**

Unistage

code	plant name	sizes
6001	VMIExxium01	14"
6002	VMIExxium02	**16, 14, 15"**
6003	VMIExxium03	**17, 15"**
6004	VMIExxium04	**16, 17, 14, 15"**
7001	vmi1Maxx	**15, 16, 18"**
7002	vmi2Maxx	**14, 18"**
7003	vmi3Maxx	**14, 15, 16"**
7004	vmi4Maxx	**15, 16, 18, 17, 14"**
7101	bj4	15"
7102	bj5	**15, 13"**
7103	bj6	**13, 15"**
7104	bj7	**15, 16"**
7105	bj9	13"
7106	bj10	**13, 15, 14"**
7201	bj8	**16, 14, 13"**
7501	us1	**12, 13"**
7502	us2	13"
7503	us3	13"

KEY READ

Adopted June config (CO limit 12, june_production_data 656,608, june_Daily_Running_Moulds). 643,928 cured / 103 diff-size COs — near-full coverage with plant-like churn. Machines concentrate on a dominant inch; amber ones flex 2-3.

645,580 cured · 98.3% coverage · 79 diff-size COs

Leaner-churn June alternative (13 machines multi-inch, amber). Higher urgency margin = fewer size switches.

Stage-1

code	plant name	sizes
6801	bj1stage1	(idle)"
6802	bj2stage1	(idle)"
6803	bj3stage1	16"
6909	nrm9stage1	12"
6911	nrm11stage1	15"
7601	ltmstage1	15"
7701	midland5stage1	15"
7801	midland1stage1	12"
7802	midland2stage1	13"
7803	midland3stage1	15"
7804	midland4stage1	15"
8001	sai1stage1	12"
8002	sai2stage1	12"
8003	sai3stage1	12"
8101	88d1stage1	(idle)"

Stage-2

code	plant name	sizes
7301	newirm	15"
8201	oldirm	12"
8301	gtic1	12"
8302	gtic2	**12, 15, 13"**
8501	vmi1	**15, 12, 13"**
8502	vmi2	**15, 16"**

Unistage

code	plant name	sizes
6001	VMIExxium01	14"
6002	VMIExxium02	**14, 15, 17"**
6003	VMIExxium03	**16, 17, 15"**
6004	VMIExxium04	**16, 17, 14"**
7001	vmi1Maxx	**15, 18"**
7002	vmi2Maxx	**15, 14, 17, 16, 18"**
7003	vmi3Maxx	**14, 16, 15"**
7004	vmi4Maxx	**16, 18, 17, 15, 14"**
7101	bj4	15"
7102	bj5	15"
7103	bj6	**13, 15"**
7104	bj7	15"
7105	bj9	13"
7106	bj10	**13, 14"**
7201	bj8	**16, 13"**
7501	us1	12"
7502	us2	13"
7503	us3	13"

KEY READ

Lower-churn June alternative: 79 diff-size COs (vs 103) — here the higher margin trims churn AND holds coverage (645,580 / 98.3%, marginally above margin=150). June's demand shape lets it switch less without losing KPI.